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Late Paleozoic Partial Remagnetization of Ordovician Rocks from Southern Oklahoma

ABSTRACT

Measurements of the NRM (natural remanent magnetization) of Lower Ordovician sediments collected from both flanks of the Arbuckle anticline (south-central Oklahoma) correspond to an upper Carboniferous–Permian magnetization direction. Directions of magnetization are well grouped and remain essentially unchanged through thermal demagnetizations up to 400°C. Correction for folding indicates that the magnetization is post-folding and that the Arbuckle orogeny was of sufficient intensity to heat and remagnetize the sediments involved.

Upon thermal demagnetization to temperatures of 550°C, the directions of magnetization changed considerably to directions distinctly removed from the observed upper Carboniferous–Permian direction. Sample intensities are very weak and therefore a reliable direction was unattainable. The data form a streaked distribution away from the Carboniferous-Permian direction. The calculated upper Carboniferous–Permian pole, corrected for the primary prefolding remanence direction observed above 400°C, is located at 128.9° E., 46.1° N. Orogenic temperature calculations based on differing stabilities of this late Paleozoic remanence on opposite flanks of the anticline indicate a more intense orogeny to the north with minimum temperatures of 200°C. Lesser temperatures, of at least 120°C but less than 200°C, prevailed on the south flank.

INTRODUCTION

Oriented samples were collected from the West Spring Creek Formation of the Ordovician Arbuckle Group on the north and south flanks of the Arbuckle anticline about 15 mi north of Ardmore, Oklahoma (97.2° W., 34.2° N). The samples were collected along the new Interstate 35 roadcut on either side of its intersection with U.S. Highway 77, as indicated by the two dots in Figure 1.

The geology of the area along the roadcut has been described by Fay (1969). The Ordovician West Spring Creek Formation consists of 1,500 ft of predominantly gray limestone. Fay has divided the formation into Upper, Middle, and Lower Zones. The Middle Zone contains many intervals of red calcic shale, red marl, red shaley limestone, and red limestone. Samples were obtained from the central part of the Middle Zone (475 ft below its upper boundary) about 760 ft below the top of the West Spring Creek Formation. The entire Paleozoic sequence in this area was folded in Late Pennsylvanian to Early Permian time (Ham and Wilson, 1967). Samples were taken from both flanks of the anticline in order to apply the fold test (Graham, 1949).

A continous stratigraphic sequence of oriented cores (spanning 80 ft) was obtained from the formation on the south flank, which dips slightly more than 50° to the southwest. Core samples were collected by means of a hand-held gasoline-powered core drill (Helsley, 1967) and were oriented with a magnetic compass. Block samples were obtained from the north flank which dips at a much steeper angle, between 65° and 75° to the northeast The dip of the strata is more variable on the more highly faulted north flank. Only a small number of samples were collected as the samples were intended as a pilot study.

Measurements were made on a PAR SM-1 spinner magnetometer. Thermal demagnetization was done in a conventional noninductively wound, twelve-sample capacity furnace. Temperature was monitored at four levels within the furnace by platinum thermocouples referenced to an ice-water solution and read on a digital voltmeter. Samples were cooled in a field-free region (± 10 gammas).

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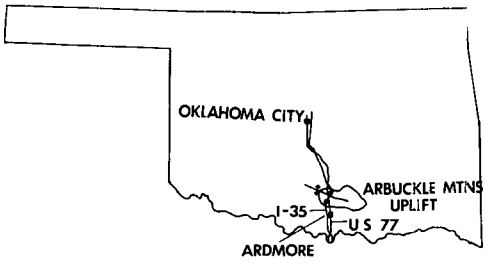


Figure 1. Location map showing the Arbuckle Mountains and Interstate 35. Sampling localities are indicated by dots on either side of the anticline axis.

NRM RESULTS

The natural remanent magnetization (NRM) directions of the samples from both the north and south flanks of the fold before structural correction were grouped around 144.3° declination and 12.6° inclination and showed slight streaking toward the present or axial geocentric dipole field direction (see Fig. 2a). Only one polarity was present. The samples are rather well grouped in NRM directions, which is indicative of their stability. The NRM directions of the few samples from the north side of the anticline show no significant difference from those of the south flank before structural correction. The mean direction of all samples is very suggestive of the upper Carboniferous–Permian (hereafter referred to as late Paleozoic) field direction.

Structurally correcting the samples to a prefold position does not significantly change the sample directions since the magnetization is directed almost along the fold axis. The slight effect structural rotation has is to move the samples from opposite flanks of the fold slightly away from each other, decreasing the grouping. Such behavior suggests a magnetization after folding (Graham, 1949).

DEMAGNETIZATION

Although postfolding remanence is strongly indicated, samples were thermally demagnetized in search of an Ordovician remanence. Pilot samples were heated to 200°C, 300°C, 400°C, and 550°C. Demagnetization above 550°C was not possible due to the weakness of the intensity of most samples at 550°C. NRM intensities of the samples are approximately bimodal and behavior during demagnetization was correlatable with the weakness of sample intensity. Upon heating, samples moved away from the present Earth's field direction and those with higher NRM intensities became well grouped at 400°C. Those with lower NRM intensities (values around .770 x 10^{-6} emu/cc) grouped moderately well at 200°C in the same direction as that of the 400°C group. These directions scattered increasingly on heating to 300°C and 400°C. Upon heating the pilot samples to 550°C, most moved markedly away from the directions held at 200° to 400°C.

As a result of the pilot experiments all samples were heated to 200°C, 400°C, and 550°C. After heating most samples to 400°C and some to 200°C, their directions formed a good group (Fig. 2d) in the late Paleozoic field direction (see Table 1).

When heated to 550°C, most samples from the south limb of the anticline moved away from the late Paleozoic direction. Samples from the north limb failed to show any change in direction at this temperature. The samples from the south limb moved in different amounts along similiar demagnetization paths. Three of these samples retained the late Paleozoic direction when heated to 550°C. As shown in Figure 2e, the result of demagnetization to 550°C is a streak of directions away from the late Paleozoic direction. It is clear that most of the sample directions have not reached stable terminal directions, and thus the end point of the streak is uncertain.

DISCUSSION

After thermal demagnetization to 400°C (and a few to 200°C), the samples from the north and south flanks have identical directions. However, since the samples from the south flank contained a primary remanence in addition to the late Paleozoic remanence, the 550°C vector was subtracted from the 400°C vector. A pole was then computed from samples from both flanks and is located at 128.9° E., 46.1° N. (dp 3.0°, dm 6.1°). This pole compares well with published upper Carboniferous and Permian pole positions for North America (Fig. 3). Removing the samples with the most deviant directions (five samples) and computing a pole (T-5) gives a pole position that is insignificantly different from the pole determined from all points (see Table 1) except for the statistics (dp 1.7°, dm 3.3°). Thus, it seems that the deviant directions are random and that the pole calculated from all of the

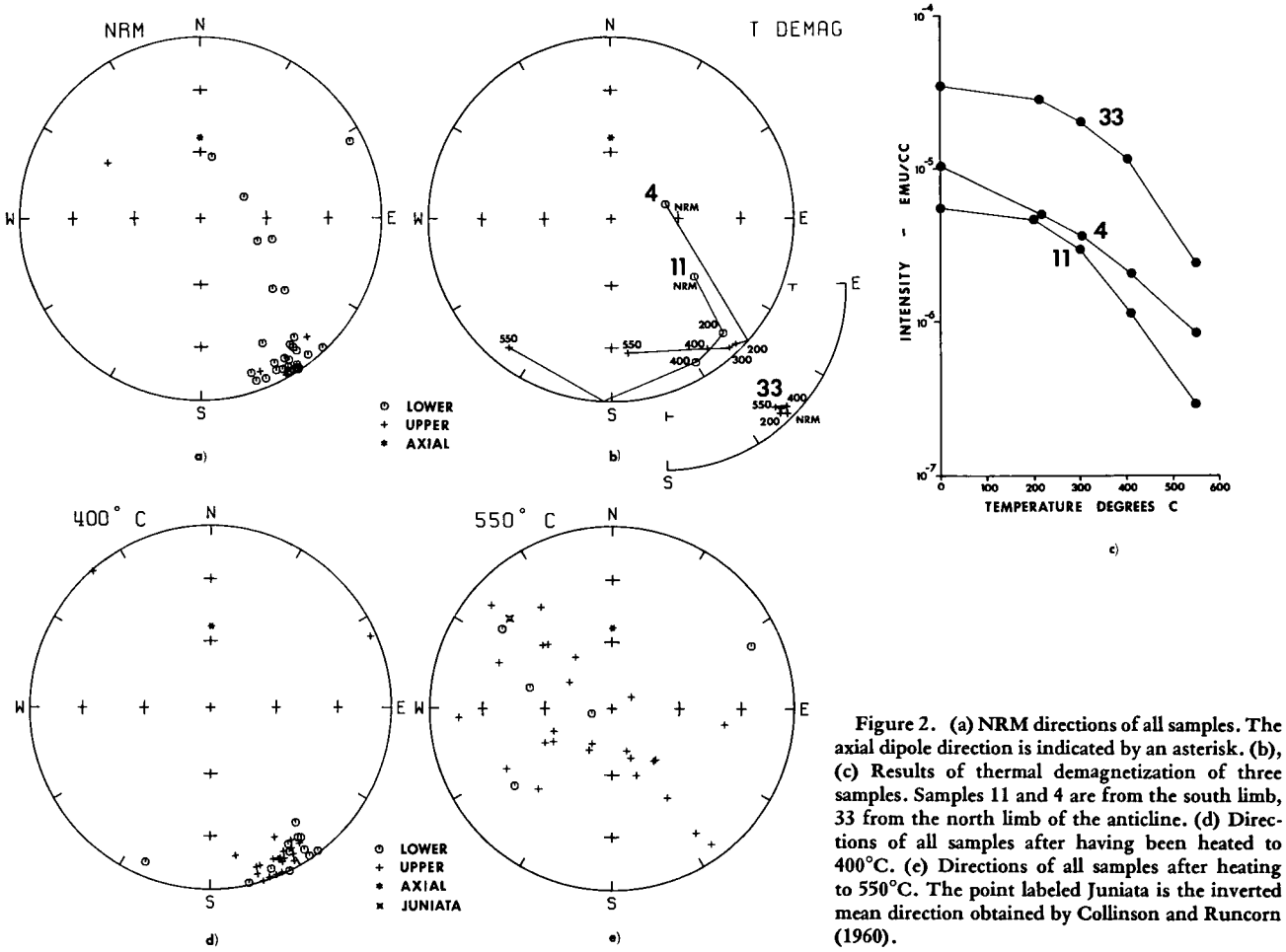


Figure 2. (a) NRM directions of all samples. The axial dipole direction is indicated by an asterisk. (b), (c) Results of thermal demagnetization of three samples. Samples 11 and 4 are from the south limb, 33 from the north limb of the anticline. (d) Directions of all samples after having been heated to 400°C. (e) Directions of all samples after heating to 550°C. The point labeled Juniata is the inverted mean direction obtained by Collinson and Runcorn (1960).

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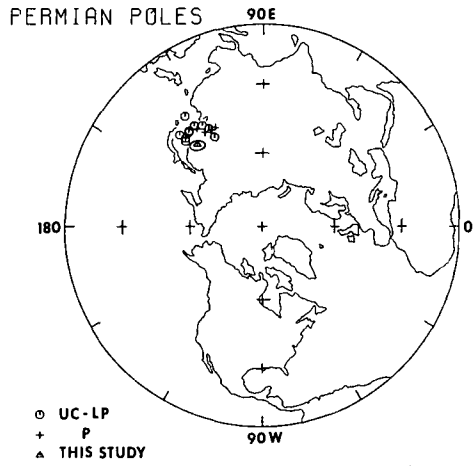


Figure 3. The 400°C pole (T-5) is shown with its oval of confidence and other published demagnetized upper Carboniferous and Permian pole positions from North America. UC-LP = upper Carboniferous to Lower Permian in age, P = Permian. Published data were selected on the basis of Irving's (1964) stability criteria from Irving, 1964; McElhinny, 1968a, 1968b, 1969; Helsley, 1971; and Gose and Helsley, 1972.

data is a good approximation to the pole position for these rocks during late Paleozoic time.

The partial thermal remanent behavior of these samples (existence of a different remanence at temperatures above 400°C) suggests that the late Paleozoic magnetization is thermal, therefore orogenic, in origin. The existence, in the West Spring Creek Formation, of red strata in the subsurface at a depth of many thousand feet (S. Pakken and L. Gatewood, 1970, oral commun.) is evidence that the remanence above 550°C may be an original characteristic of the rock, that is, Ordovician remanence.

A previous study of Ordovician rocks was done by Collinson and Runcorn (1960) on the Juniata Formation of Pennsylvania. The inverse of their mean direction is plotted in Figure 2e. It lies near the end in the streak formed by the West Spring Creek directions, suggesting that the remanence observed in these rocks may truly be Ordovician.

OROGENIC TEMPERATURE

The differing demagnetization results at higher or lower temperature depending on NRM intensities may reflect differing blocking diameters due to varying orogenic temperatures. Since some samples have only a very stable late Paleozoic magnetization, it seems that some parts of the Arbuckle anticline were either heated hotter than others or remained hot longer. The fact that the samples from the north limb showed only late Paleozoic magnetization while those from the south limb showed a hard component predating this magnetization suggests that the intensity of the orogeny was greater to the north. This observation is in agreement with field evidence. Abundant faults and an overturned fold on the north limb in the sampling area in contrast to fewer faults and a simple homoclinal dip on the south limb indicates greater deformation to the north.

The thermal demagnetization experiments established the fact that a few samples, particularly those from the north flank, retain a stable late Paleozoic magnetization direction after heating to 550°C for approximately 10 min, while late Paleozoic magnetization in samples from the south is not stable at 550°. Using this experimental fact and a relation (and accompanying assumptions) derived by Stacy (1963), an indication of a minimum orogenic temperature can be calculated. If, in the event that the orogeny was a two-stage one as suggested by Ham and Wilson (1967) from a study of the orogenic conglomerates, the temperature calculations apply to the last orogenic pulse. Calculations were made for an orogeny whose maximum thermal event lasted 1 to 10 m.y. These calculations yield a minimum temperature estimate for the north flank samples and some samples from the south flank of at least 200°C, the exact value depending upon assumptions of detailed magnetic behavior and duration of the orogeny. In the samples from the south flank, late Paleozoic magnetization is stable to 400°C or somewhat higher temperatures but disappears by 550°C. The south flank must, therefore, have been heated to at least 120°C but less than 200°C for one to ten m.y. Thus, in the vicinity of the north flank of the Arbuckle anticline, temperatures were 200°C or more, while on the south flank temperatures did not exceed 120° to 200°C.

The temperatures may have been higher if the orogenic event was shorter than 1 m.y. or if the magnetic stability persists for times in excess of ten min in the laboratory or, for the north flank strata, persists to temperatures in excess of 550°C. However, for an orogeny of 10,000 yrs and a laboratory stability of the

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TABLE 1. WEST SPRING CREEK LATE PALEOZOIC MAGNETIZATION DIRECTIONS

	NRM, 400°C										
	D	I	N	R	K	α_{95}	Del	Long, E.	Lat, N.	Dp	Dm
NRM	144.3	12.6	34	29.66	7.6	9.6	28.9	129.5	37.2	5.0	9.7
All samples 400°C	149.4	-1.0	35	32.99	17.0	6.1	19.4	129.7	45.8	3.0	6.1
400°C, Most deviant samples removed (5)	150.2	-1.9	29	28.57	65.9	3.3	9.8	129.2	46.6	1.7	3.3

magnetization of 2 hrs, the minimum temperature for the north does not exceed 300°C. The lack of metamorphism in this area and the abundance of faulting may be indicators of a low-temperature orogeny consistent with the paleomagnetic temperature evidence.

CONCLUSIONS

The pole position from the folded West Spring Creek Formation agrees fairly well with previously published Permian and upper Carboniferous pole positions from North America (Fig. 3). The close agreement of this pole with the published ones and the occurrence of the same magnetization direction on both limbs of the fold indicates that the Arbuckle orogeny was of sufficient intensity to remagnetize these sediments by heating to, in some places, at least 200°C. However, orogenic temperatures on the edge of the uplift area (south flank) were less (120° to 200°C) since a previous magnetization direction (possibly Ordovician) is retained by those rocks.

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